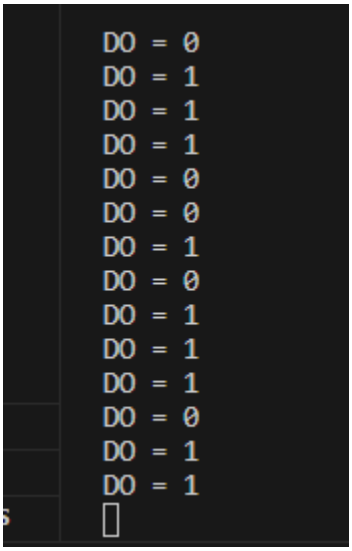
Josiah Schmitz

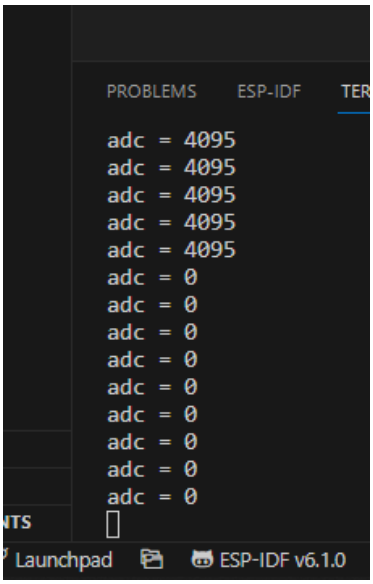
Embedded Systems

Lab 6: GPIO and ADC

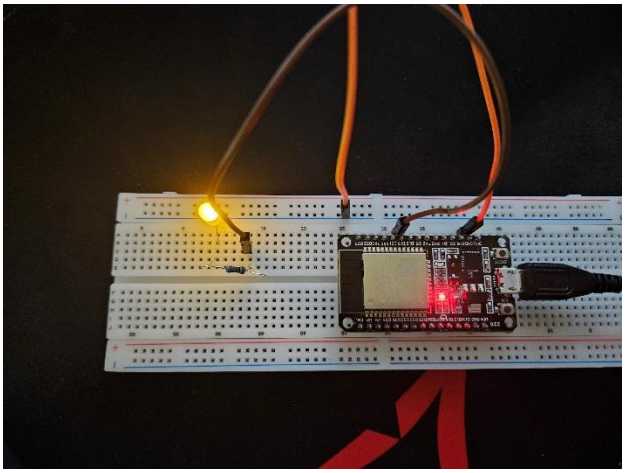
Experiment B Output



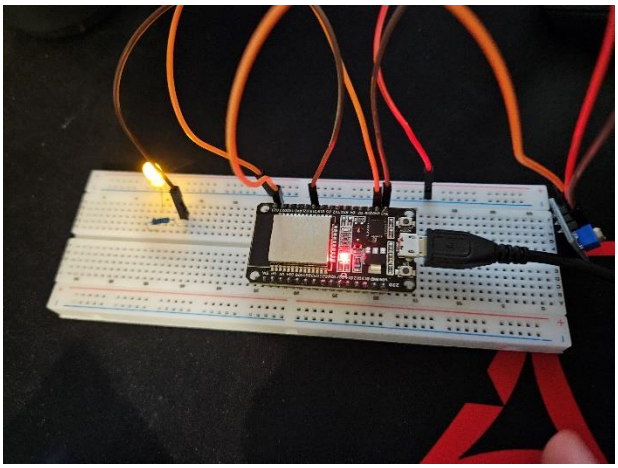
Experiment C Output



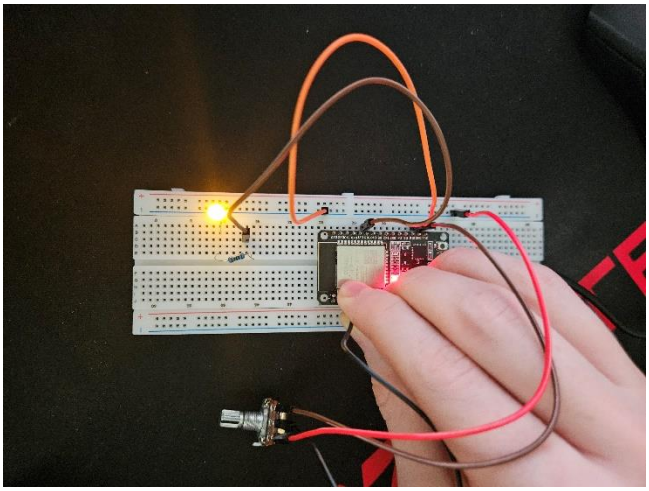
Experiment A:



Experiment B:



Experiment C:



Observations

Experiment A: The LED turned off for a half second, then turned on for a half second in a continuous loop. The serial monitor looped between displaying “LED ON” and “LED OFF”

Experiment B: The LED turned on when the sensor was covered and turned off when the sensor was uncovered. D0 was 0 when the LED was turned on, and D0 was 1 when the LED was turned off.

Experiment C: The smallest threshold was 0, and the largest threshold was 4095 (as seen above). I left the threshold set to the original 2000, so the LED turned on only when the ADC value was greater than 2000.

Critical Thinking Questions:

1. I had to deal with the ESP32 not sitting snugly in the board, causing the pin connections to randomly become loose and disconnect when I was trying to get readings. I also noticed a small delay between when the LDR sensor was covered and the LED turning on. This never happened when using Wokwi.
2. The system will reset because it’s worried that lack of free CPU time means it’s frozen. Watchdog Timer is what reacts to a delay-less loop.
3. The LDR uses digital input while the potentiometer uses analog input. Digital input simply provides a 0 or 1 to the system while analog input provides a predetermined range of values. The potentiometer controls how sensitive the sensor is to light.
4. Because 4095 represents the maximum voltage of 3.3V provided by the ESP32. A reading of 2048 is essentially the halfway point of that maximum voltage.
5. Any input marked as ADC on the ESP32 PINOUT chart, such as GPIO34, GPIO39, or GPIO36. Because the LED is an output, and the sensor is an input.

6. Download mode is needed so the system knows it is supposed to be receiving a new program from the computer.
7. I simply used the default threshold, which seemed to work well. If the threshold were set close to the normal reading, the light would constantly fluctuate between on and off.